Waves in Supersonic Flow

AA 504 (Compressible Fluid Mechanics) Autumn 2026, UW

Assignment: Homework #4

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Date: 4NOV2026

Location: Suzzallo library, UW, Seattle, WA

Table of Contents

Problem 1. Shock Wave Propagation 1
Problem Statement 1
Analysis 1
Problem 3. Acoustic Wave Propagation 5
Problem Statement 5
Analysis 5
Problem 3. Pressure Fields and Fluid Acceleration 5
Problem Statement 5
Statement 5
References 6

```
clear % Clears workspace
clc % Clears command window
close all % Closes all figures
```

Problem 1. Shock Wave Propagation

Problem Statement

A Mach 3 shock wave propagates to the right along a shock tube and then collides with the closed end of the tube. The initial pressure of the gas is p1. Draw the x-t diagram. What is the pressure on the end of the tube after the wave arrives? (*Hint:* The boundary condition of zero speed at the end of the tube can be satisfied by adding a virtual wave of identical strength propagating to the left.)

Analysis

[*pending*]

x-t diagram:

```
%% Mach 3 Shock Reflection - x-t Diagram
% Incident shock travels right and reflects from a rigid wall.
% Horizontal axis: position x
% Vertical axis: time t
%
% Assumptions:
%   Air, gamma = 1.4
%   Initially stationary gas
%   Incident shock Mach number Ms = 3

clear; clc; close all;

%% 1. Gas and shock properties
gamma = 1.4;
Ms = 3;

a1 = 343;           % Initial speed of sound [m/s]
L = 1.0;            % Tube length [m]

% Incident shock speed (laboratory frame)
Vs = Ms*a1;

% State 2: behind incident shock
p2_p1 = 1 + 2*gamma/(gamma+1)*(Ms^2-1);

rho2_rho1 = ((gamma+1)*Ms^2)/ ...
            ((gamma-1)*Ms^2+2);

u2 = Vs*(1-1/rho2_rho1);

% Speed of sound behind incident shock
a2 = a1*sqrt(p2_p1/rho2_rho1);

%% 2. Reflected shock properties
% Reflected shock must bring the gas to rest: u3 = 0

K = (gamma+1)*u2/(2*a2);

% Solve MR - 1/MR = K
MR = (K + sqrt(K^2+4))/2;

% Reflected shock speed in laboratory frame
Vr = u2 - MR*a2;     % Negative = moving left

% Pressure behind reflected shock
p3_p2 = 1 + 2*gamma/(gamma+1)*(MR^2-1);
p3_p1 = p2_p1*p3_p2;

%% 3. Shock trajectories
% Incident shock begins at x = 0, t = 0
```

```
% Reflection occurs at x = L

tR = L/Vs;           % Time shock reaches wall
tEnd = 1.8*tR;       % Plot beyond reflection

tIncident = linspace(0,tR,200);
xIncident = Vs*tIncident;

tReflected = linspace(tR,tEnd,200);
xReflected = L + Vr*(tReflected-tR);

%% 4. Plot x-t diagram
figure('Color','w','Position',[150 100 850 650]);
hold on; box on;

% Incident shock
h1 = plot(xIncident,1e3*tIncident, ...
    'LineWidth',2.8,'Color',[0.90 0.42 0.12]);

% Reflected shock
h2 = plot(xReflected,1e3*tReflected, ...
    'LineWidth',2.8,'Color',[0.10 0.50 0.72]);

% Closed wall
h3 = xline(L,'k--','LineWidth',1.8);

% Reflection point
h4 = plot(L,1e3*tR,'ko', ...
    'MarkerFaceColor','k','MarkerSize',7);

%% 5. Region labels
text(0.75*L,0.35*1e3*tR, ...
    {'State 1','p_1, u_1 = 0'}, ...
    'HorizontalAlignment','center','FontSize',12);

text(0.30*L,0.90*1e3*tR, ...
    {'State 2','p_2 = 10.33p_1'}, ...
    'HorizontalAlignment','center','FontSize',12);

text(0.85*L,1.48*1e3*tR, ...
    {'State 3','p_3 = 51.67p_1','u_3 = 0'}, ...
    'HorizontalAlignment','center','FontSize',12);

% Reflection annotation
text(L-0.03,1e3*tR+0.04, ...
    'Reflection', ...
    'HorizontalAlignment','right', ...
    'FontSize',11,'FontWeight','bold');

%% 6. Format figure
```

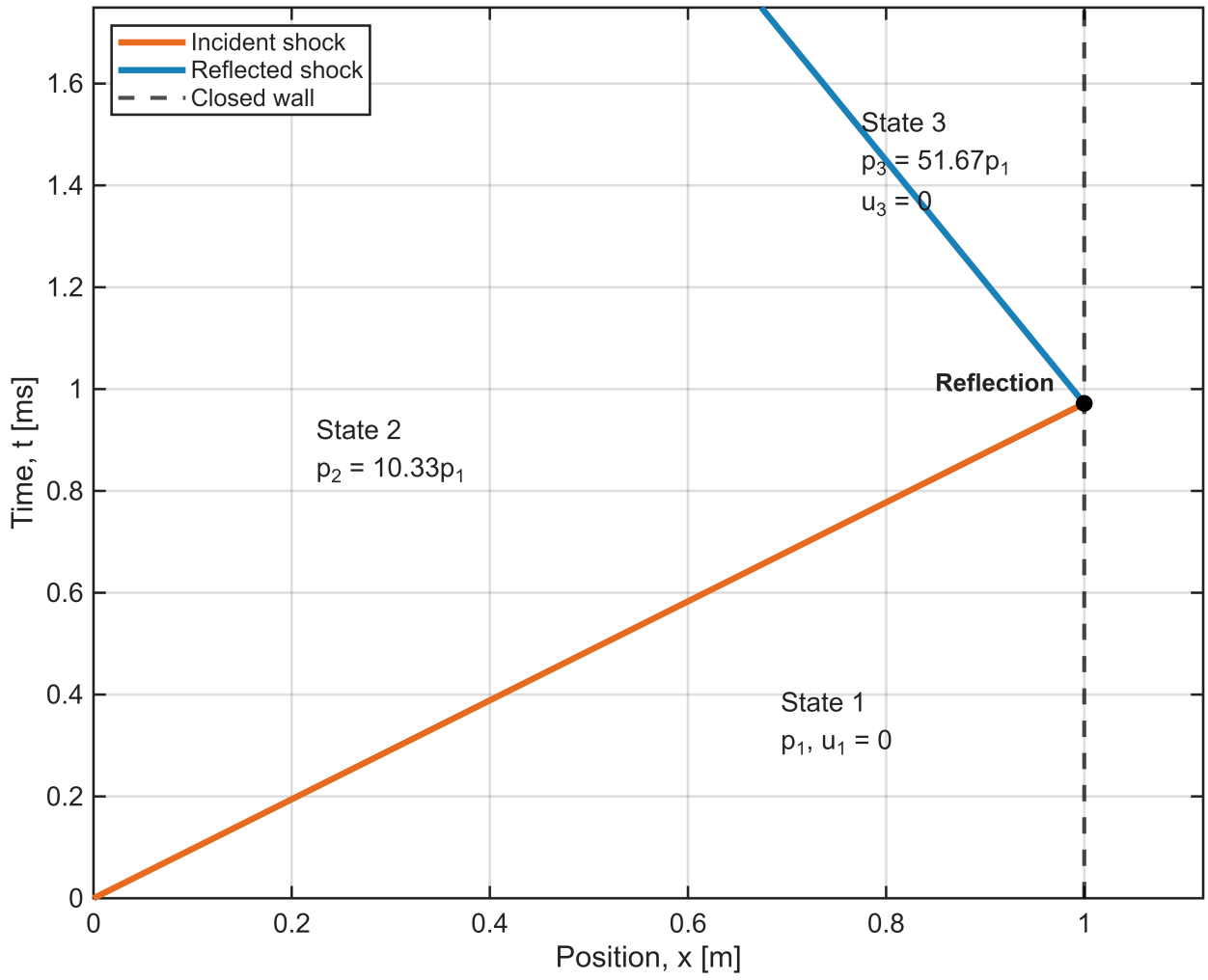
```
xlabel('Position, x [m]','FontSize',13);
ylabel('Time, t [ms]','FontSize',13);

title('Mach 3 Shock Reflection at a Closed Wall', ...
    'FontSize',15,'FontWeight','bold');

legend([h1 h2 h3], ...
    {'Incident shock','Reflected shock','Closed wall'}, ...
    'Location','northwest');

xlim([0 1.12*L]);
ylim([0 1e3*tEnd]);

grid on;
set(gca,'FontSize',12,'LineWidth',1.1, ...
    'Color','w','GridAlpha',0.15);
```




```
%% 7. Print results
```

```
fprintf('\n--- Shock Reflection Results ---\n');
```

```
--- Shock Reflection Results ---
```

```
fprintf('Incident shock speed: %.2f m/s\n',Vs);
```

```
Incident shock speed: 1029.00 m/s
```

```
fprintf('Reflected shock speed: %.2f m/s\n',Vr);
```

```
Reflected shock speed: -419.22 m/s
```

```
fprintf('Reflection time:       %.4f ms\n',1e3*tR);
```

```
Reflection time:       0.9718 ms
```

```
fprintf('Incident pressure:     %.4f p1\n',p2_p1);
```

```
Incident pressure:     10.3333 p1
```

```
fprintf('Wall pressure:         %.4f p1\n',p3_p1);
```

```
Wall pressure:         51.6667 p1
```

```
%% 8. Optional PDF export
% exportgraphics(gcf,'Mach3_Shock_Reflection.pdf', ...
%     'ContentType','vector');
```

Problem 3. Acoustic Wave Propagation

Problem Statement

Suppose that the initial wave in Problem (1) is instead acoustic, of small pressure amplitude . What is the pressure on the end wall after the wave arrives? Explain.

Analysis

[*pending*]

Problem 3. Pressure Fields and Fluid Acceleration

Problem Statement

Based off of the movie "Pressure Fields and Fluid Acceleration" [3], write one paragraph aimed at a "bright high school kid" that explains one aspect of physics illustrated in the movie.

Statement

Imagine water flowing toward the nose of an object (a submarine, or a ship!): most of the flow turns and flows around the sides, but right at the nose, the water comes to a stop. As it slows down, its pressure rises.

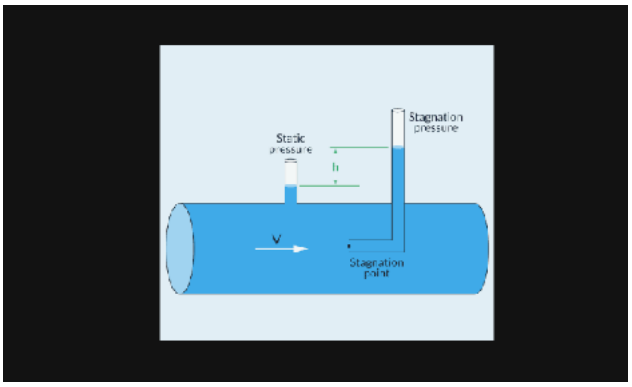
Stagnation pressure is the pressure a moving fluid would reach if it were brought to rest (to a complete stop!) without losses from friction or other irreversible effects.

In the movie's nearly frictionless water experiments, it equals the fluid's **static** (or "ordinary") **pressure** plus its **dynamic pressure**, ($\frac{1}{2}\rho u^2$), which represents its kinetic energy per unit volume.

$$p + \frac{1}{2}\rho u^2 = p_{stagnation}$$

As water speeds up through a contraction, its static pressure falls, but its stagnation pressure stays approximately constant. A Pitot tube measures stagnation pressure by bringing the fluid to rest at its forward-facing opening; comparing that reading with the static pressure lets us calculate the flow speed.

```
imshow("pitot_tube.png");
```



References

[1] Liepmann, H. W., & Roshko, A. (2001). *Elements of gasdynamics*. Dover Publications. (Original work published 1957)

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] YouTube '6. Pressure Fields and Fluid Acceleration'.